EE 121B: Principles of Semiconductor Device Design

Lecture 12:

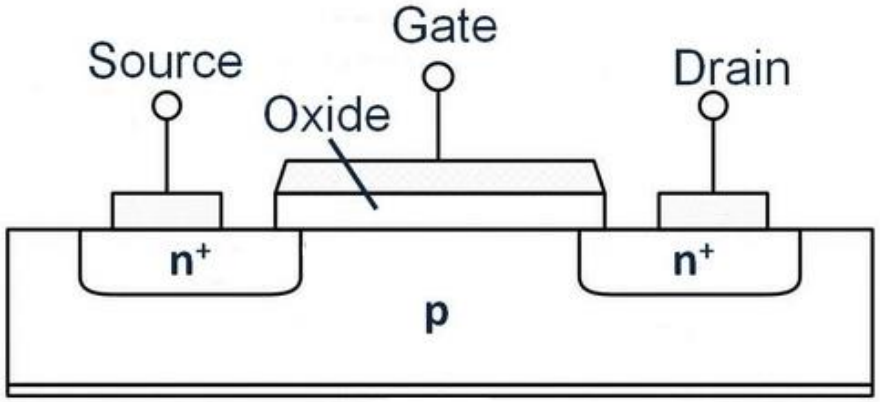
The MOS Capacitor (continued)

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. Metal-Semiconductor Contacts
3. P-N Junction
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

Lecture 12

OUTLINE

- The MOS Capacitor (cont'd)
  - Electrostatics
  - Small-signal capacitance
    - (C-V characteristics)

Reading: Pierret 16.3, 16.4; Hu 5.2-5.6

Recall: Voltage Drops in the MOS System

- In general,

Voltage dropped within oxide

Voltage dropped within Si

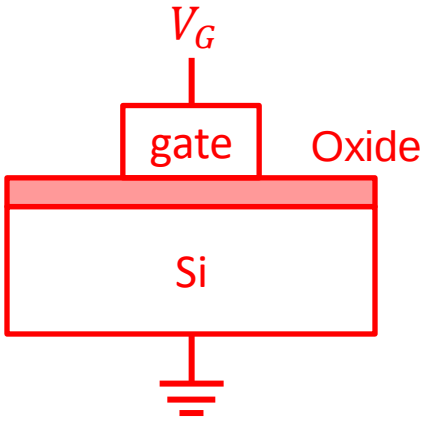
$$V_G = V_{FB} + V_{ox} + \phi_s$$

Flat band voltage
(analogous to built-in potential)

where

$$qV_{FB} = \Phi_{MS} = \Phi_M - \Phi_S$$

V_{ox} is the voltage dropped across the oxide
(V_{ox} = total amount of band bending in the oxide)



ϕ_s is the voltage dropped in the silicon
(total amount of band bending in the silicon)

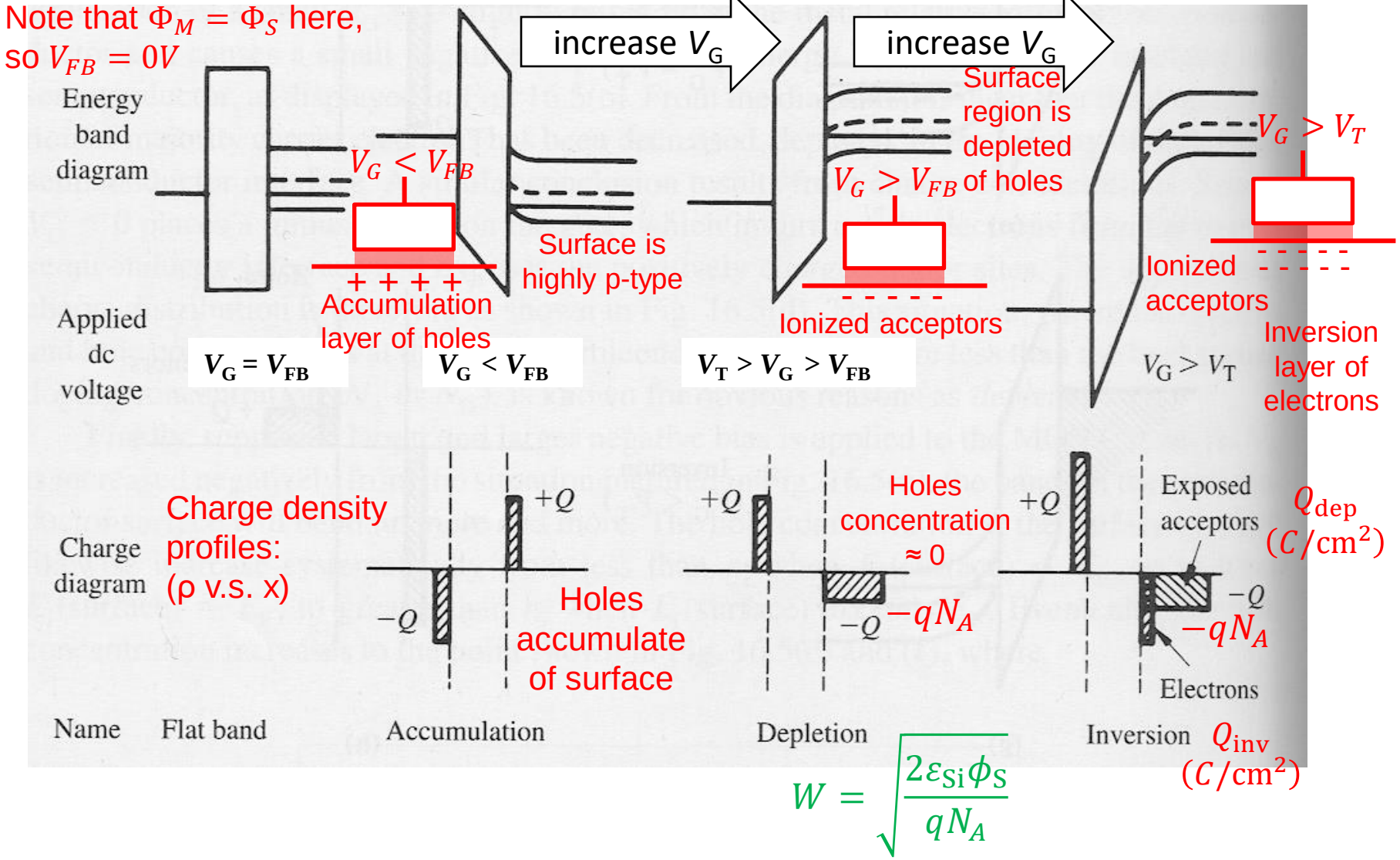
$$q\phi_S = E_i(bulk) - E_i(surface)$$

- For example:

When $V_G = V_{FB}$, $V_{ox} = \phi_s = 0$, *i.e.* there is no band bending

Recall: MOS Operating Regions (p-type Si)

(NMOS device)

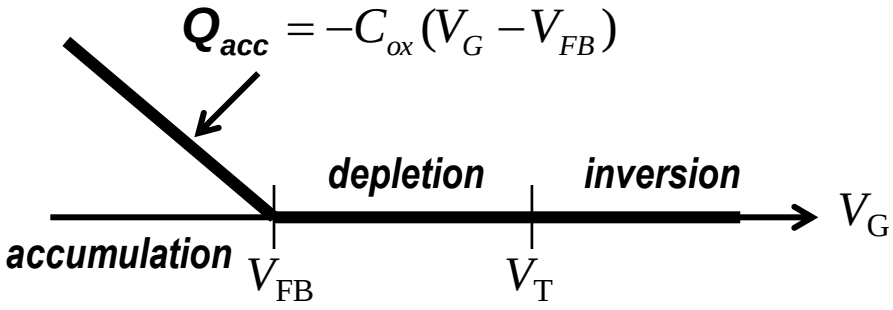


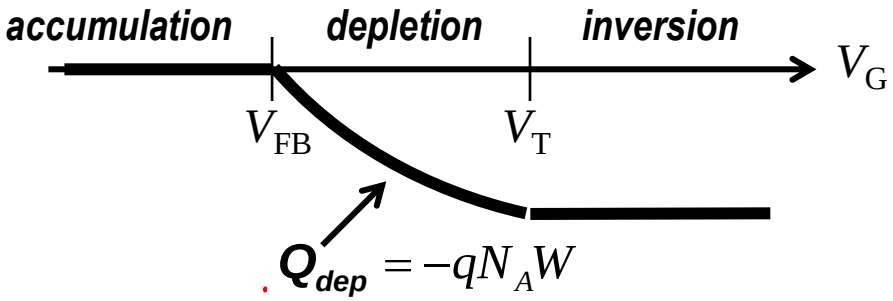
$$W = \sqrt{\frac{2\varepsilon_{Si}\phi_S}{qN_A}}$$

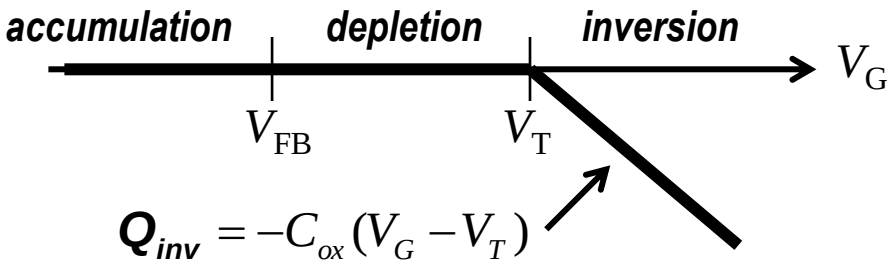
R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 16.6

Spoiler Alert!

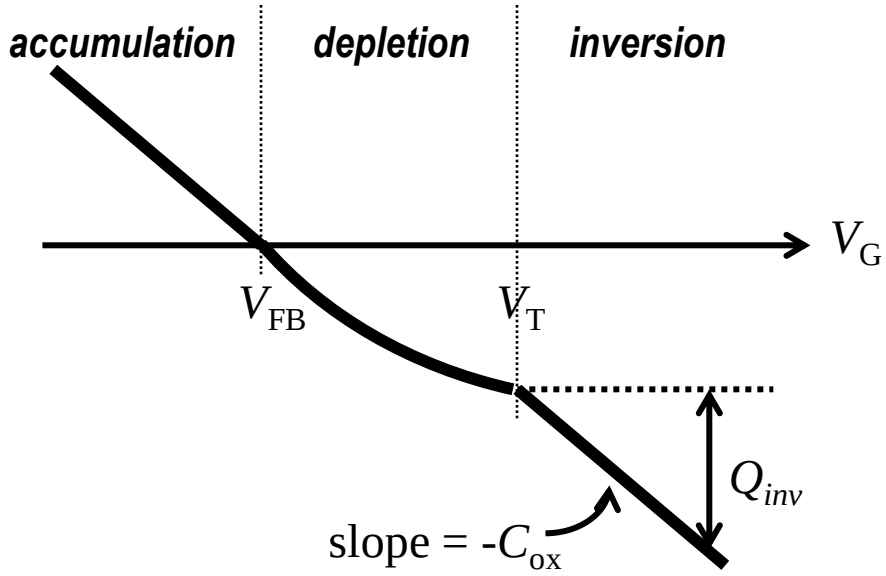
Total Charge Density in Si, Q_s (p-type Si)





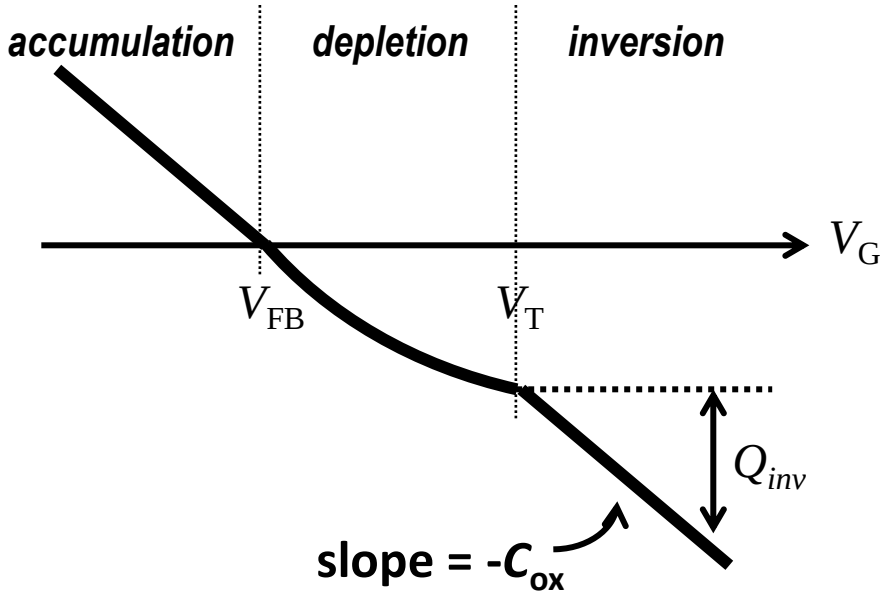


$$Q_s = Q_{acc} + Q_{dep} + Q_{inv}$$



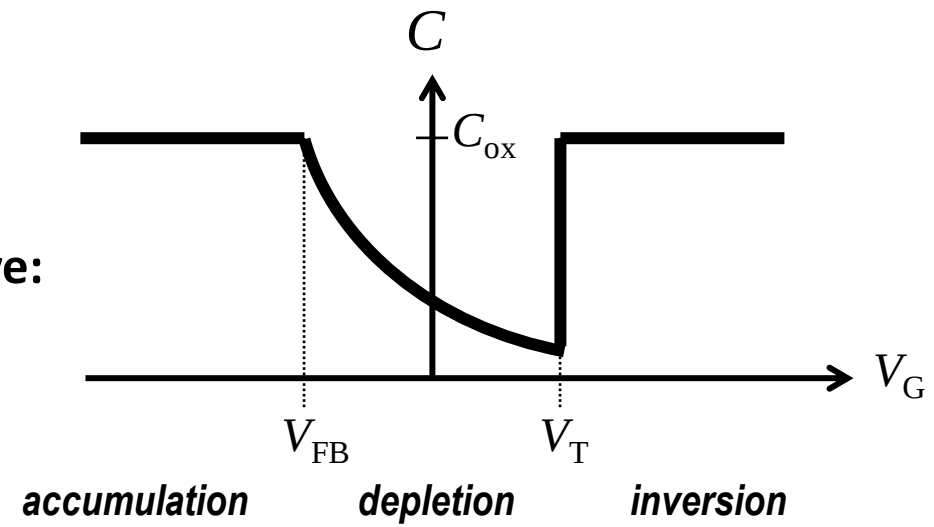
Spoiler Alert!

MOS *C-V* Characteristics (p-type Si)



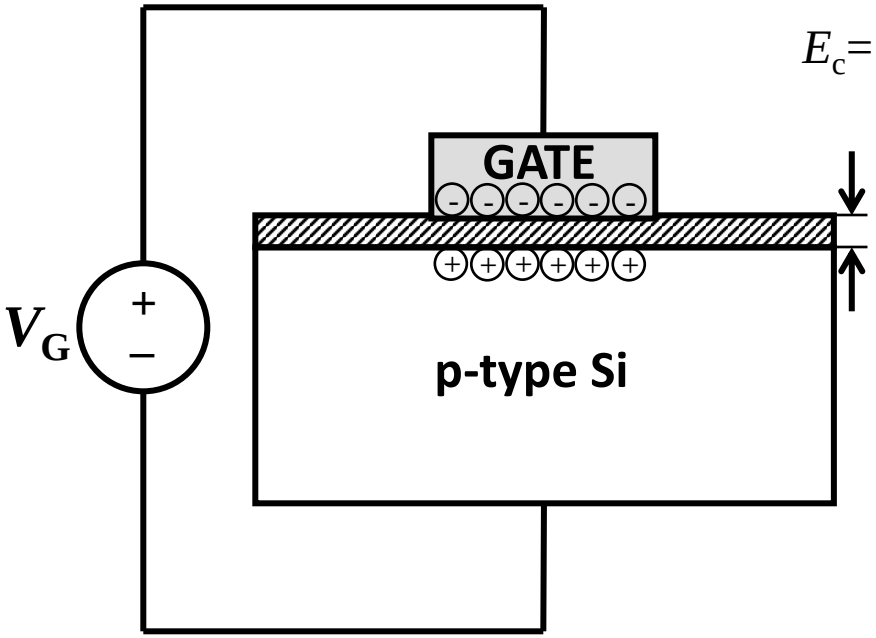
$$C = \left| \frac{dQ_s}{dV_G} \right|$$

Ideal *C-V* curve:

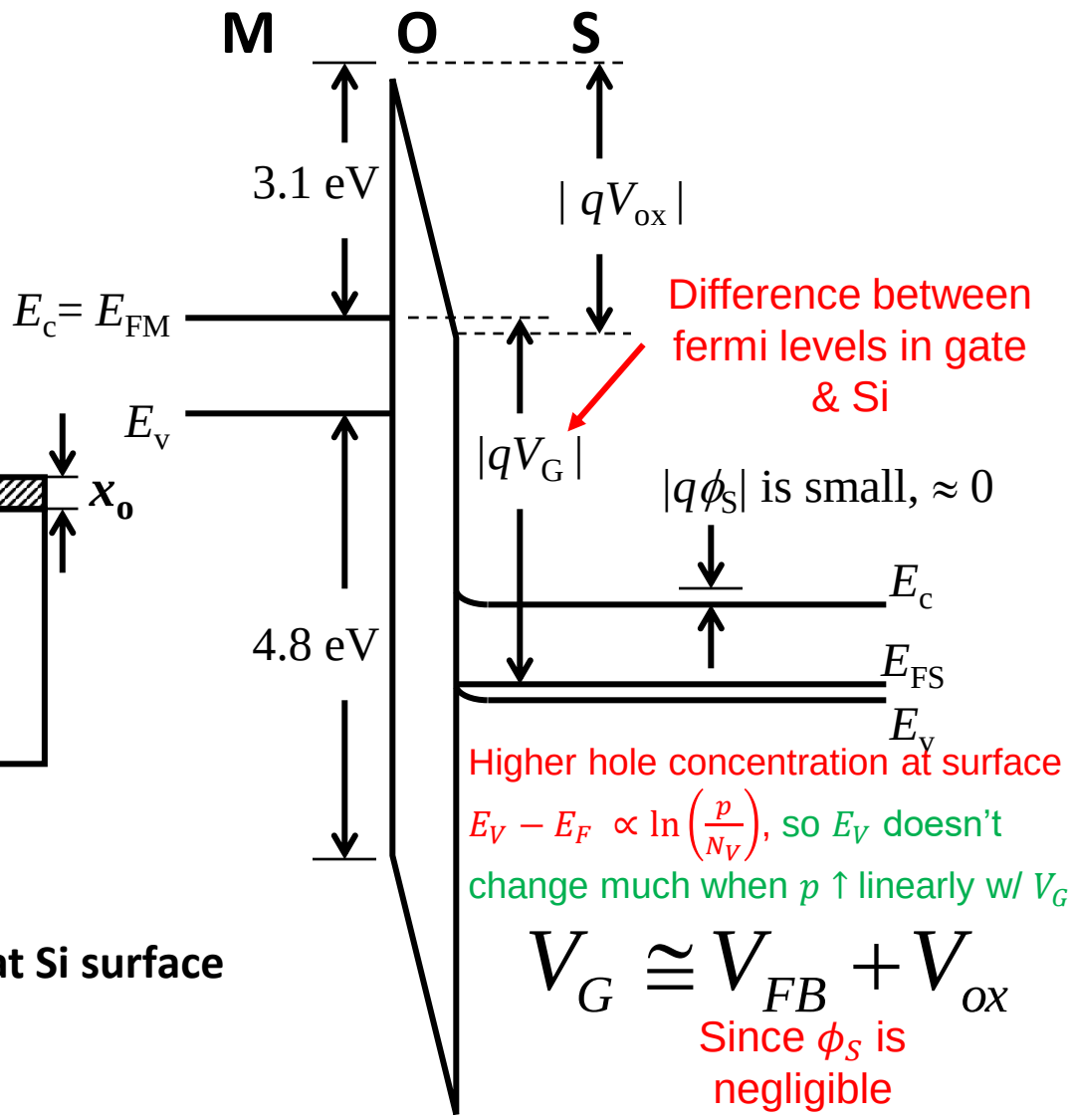


Accumulation
(n+ poly-Si gate, p-type Si)

$V_G < V_{FB}$



Mobile carriers (holes) accumulate at Si surface



Difference between fermi levels in gate & Si

Higher hole concentration at surface

$E_V - E_F \propto \ln\left(\frac{p}{N_V}\right)$, so E_V doesn't change much when $p \uparrow$ linearly w/ V_G

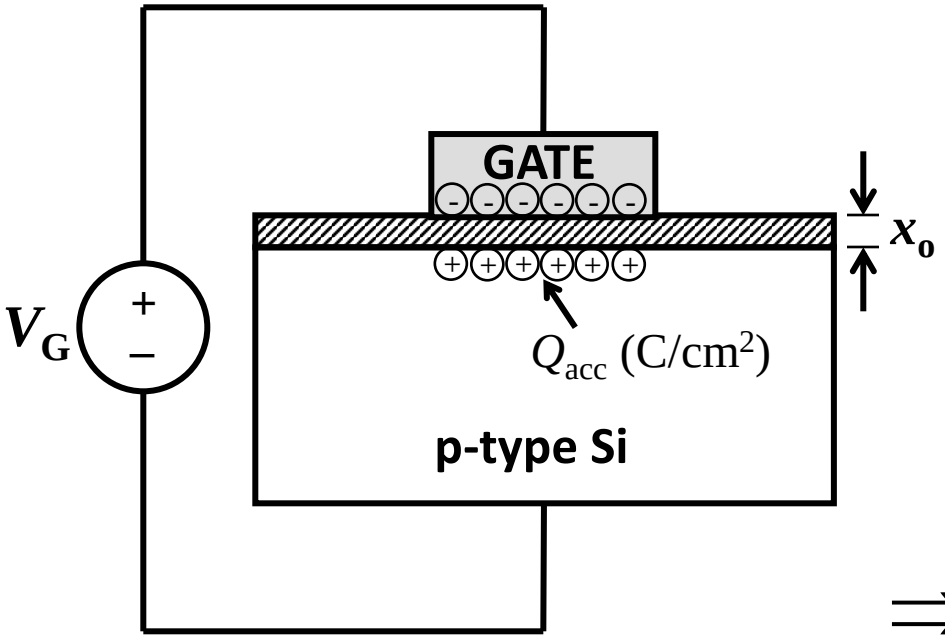
$$V_G \cong V_{FB} + V_{ox}$$

Since ϕ_S is negligible

Accumulation Layer Charge Density

$V_G < V_{FB}$

$$V_{ox} \cong V_G - V_{FB}$$



From Gauss' Law:

$$\mathcal{E}_{ox} = -Q_{acc} / \varepsilon_{SiO_2}$$

per area

$$V_{ox} = \mathcal{E}_{ox} x_o = -Q_{acc} / C_{ox}$$

per area

where $C_{ox} \equiv \varepsilon_{SiO_2} / x_o$

(units: F/cm^2)

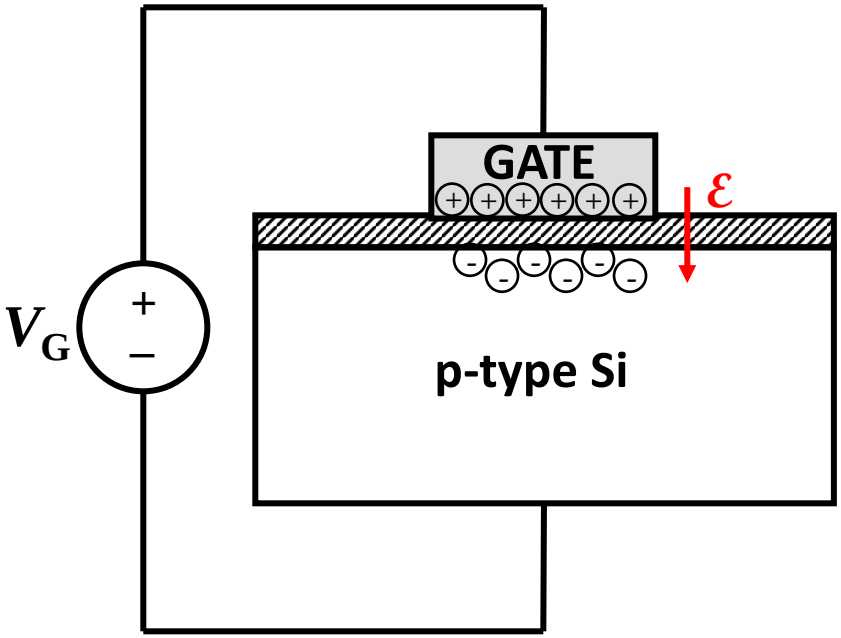
$$\Rightarrow \boxed{Q_{acc} = -C_{ox}(V_G - V_{FB})} > 0$$

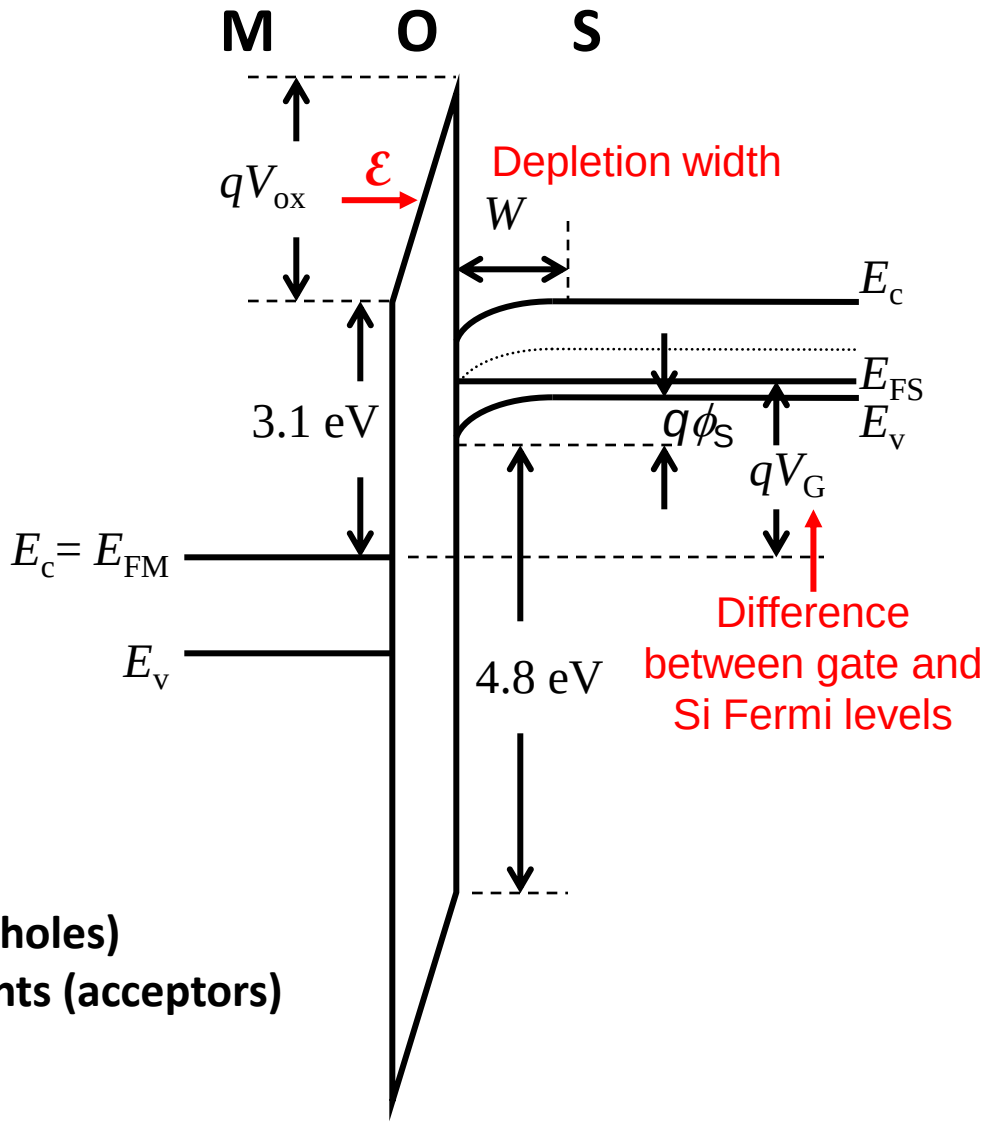
Units: C/cm^2

$$|Q| = C|V| = \varepsilon \frac{A}{d}|V| \rightarrow \frac{Q}{A} = |V|\frac{\varepsilon}{d} \rightarrow |Q|_{per\ area} = |V| C_{per\ area}$$

Depletion
(n+ poly-Si gate, p-type Si)

$V_T > V_G > V_{FB}$





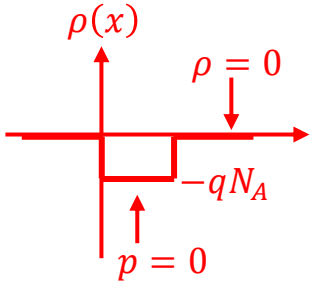
Si surface is depleted of mobile carriers (holes)

=> Surface charge is due to ionized dopants (acceptors)

Depletion Width *W* (p-type Si)

- **Depletion Approximation:**
  The surface of the Si is depleted of mobile carriers to a depth *W*.
- **The charge density within the depletion region is**

$$\rho \cong -qN_A \qquad (0 \le x \le W)$$



- **Poisson's equation:**

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_{Si}} \cong -\frac{qN_A}{\varepsilon_{Si}} \qquad (0 \le x \le W)$$

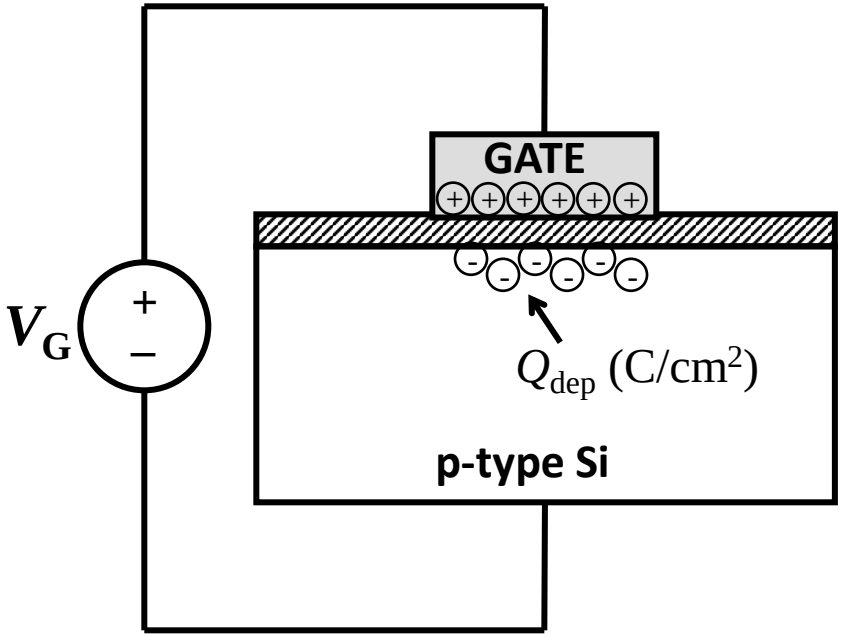
- **Integrate twice, to obtain ϕ_S:**

Voltage dropped across the silicon depletion region

$$\phi_S = \frac{qN_A}{2\varepsilon_{Si}} W^2 \quad \Rightarrow \boxed{W = \sqrt{\frac{2\varepsilon_{Si}\phi_S}{qN_A}}}$$

To find ϕ_s for a given V_G, we need to consider the voltage drops in the MOS system…

Voltage Drops in Depletion (p-type Si)



From Gauss' Law:

$$\mathcal{E}_{ox} = -Q_{dep} / \varepsilon_{SiO_2}$$

$$V_{ox} = \mathcal{E}_{ox} x_o = -Q_{dep} / C_{ox}$$

***Q_{dep}* is the integrated charge density in the Si:**

$$Q_{dep} = -qN_A W = -\sqrt{2qN_A \varepsilon_{Si} \phi_S}$$

$$W = \sqrt{\frac{2\varepsilon_{Si}\phi_S}{qN_A}}$$

$$|V_{ox}| = \frac{|Q_{dep}|}{C_{ox}}$$

$$V_G = V_{FB} + \phi_S + V_{ox} = V_{FB} + \phi_S + \frac{\sqrt{2qN_A \varepsilon_{si} \phi_S}}{C_{ox}}$$

Surface Potential in Depletion (p-type Si)

$$V_G = V_{FB} + \phi_S + \frac{\sqrt{2qN_A\varepsilon_{si}\phi_S}}{C_{ox}}$$

- **Solving for ϕ_S, we have**

$$\sqrt{\phi_S} = \frac{\sqrt{qN_A\varepsilon_{si}}}{\sqrt{2}C_{ox}}\left[\sqrt{1+\frac{2C_{ox}{}^2(V_G - V_{FB})}{qN_A\varepsilon_{si}}} - 1\right]$$

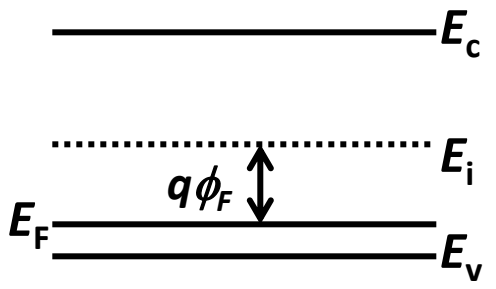
$$\phi_S = \frac{qN_A\varepsilon_{si}}{2C_{ox}{}^2}\left[\sqrt{1+\frac{2C_{ox}{}^2(V_G - V_{FB})}{qN_A\varepsilon_{si}}} - 1\right]^2$$

Bulk Semiconductor Potential, ϕ_F

$$q\phi_F \equiv E_i(bulk) - E_F$$

- p-type Si: $p = n_i e^{(E_i - E_F)/kT} \cong N_A$

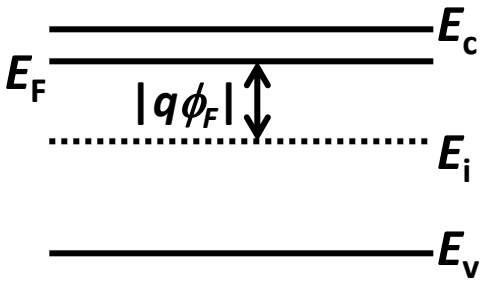
$$\phi_F = \frac{kT}{q}\ln(N_A / n_i) > 0$$



Example: $N_A = 10^{18}\ \text{cm}^{-3} \rightarrow \phi_F = 0.48\ \text{eV}$

- n-type Si: $n = n_i e^{(E_F - E_i)/kT} \cong N_D$

$$\phi_F = -\frac{kT}{q}\ln(N_D / n_i) < 0$$



Threshold Condition ($V_G = V_T$)

- **When V_G is increased to the point where ϕ_s reaches $2\phi_F$, the surface is said to be strongly inverted. This is the threshold condition.**

$$V_G = V_T \rightarrow \phi_S = 2\phi_F$$
$$E_i(bulk) - E_i(surface) = 2[E_i(bulk) - E_F]$$
$$E_F - E_i(surface) = E_i(bulk) - E_F$$
$$\rightarrow n_{surface} = N_A$$

$n = n_i e^{(E_F - E_i)/kT}$

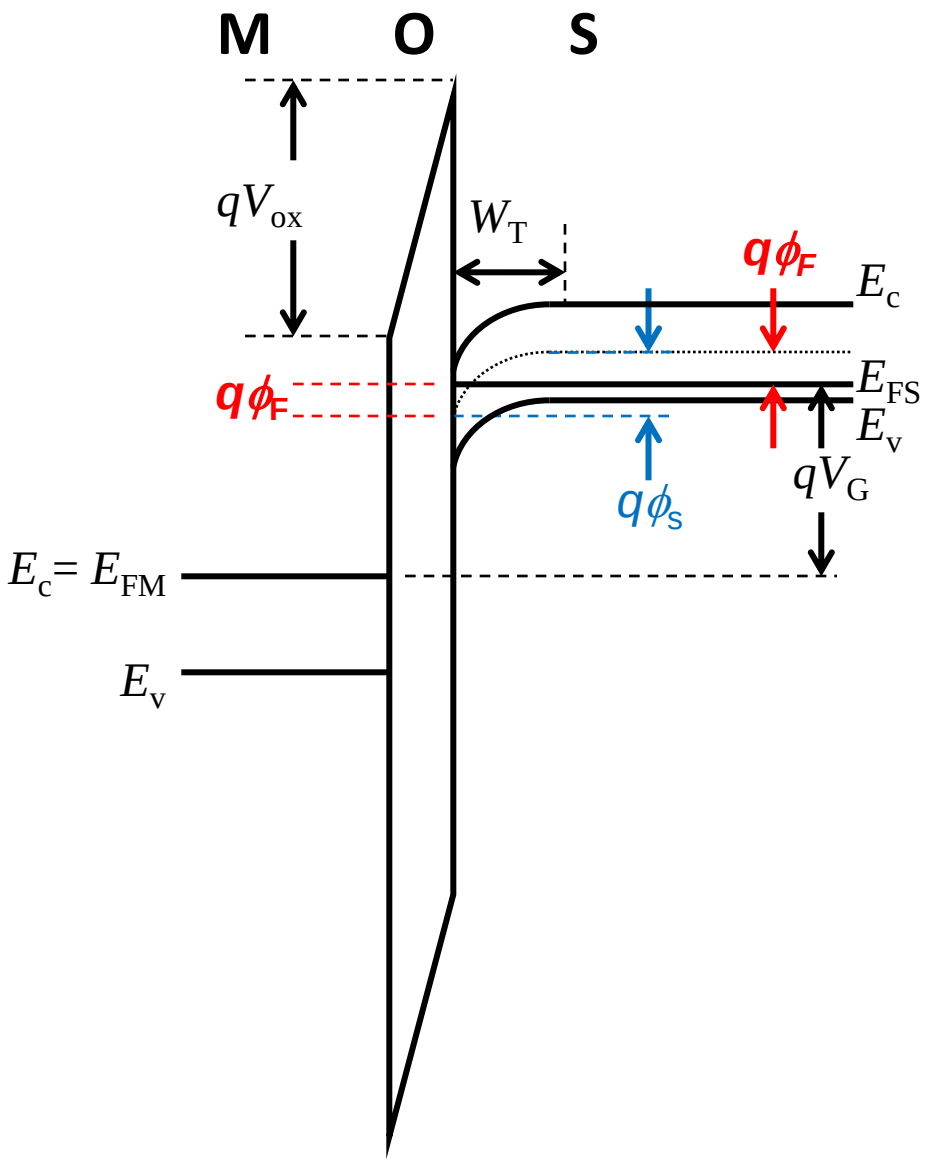
$p = n_i e^{(E_i - E_F)/kT}$

(The surface is n-type to the same degree as the bulk is p-type.)

MOS Band Diagram at Threshold (p-type Si)

$$\phi_S = 2\phi_F = 2\frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right)$$

$$W = W_T = \sqrt{\frac{2\varepsilon_{Si}(2\phi_F)}{qN_A}}$$



Threshold Voltage

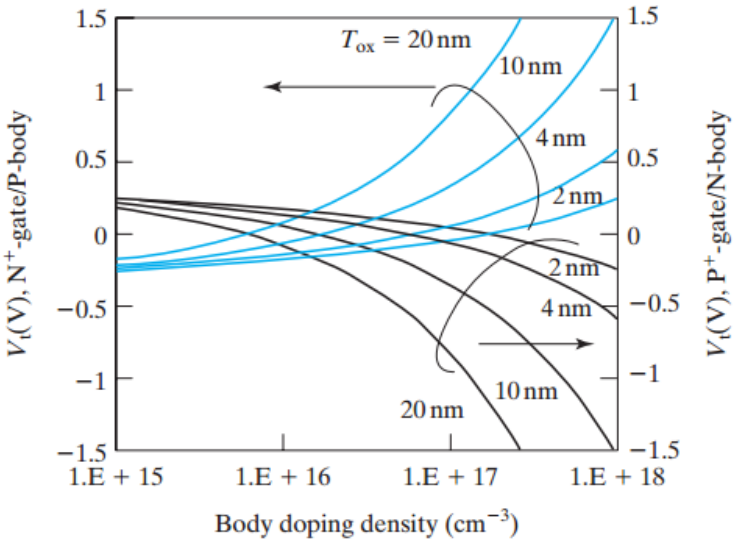
- **For p-type Si:**

$$V_G = V_{FB} + \phi_S + V_{ox} = V_{FB} + \phi_S + \frac{\sqrt{2qN_A\varepsilon_{si}\phi_S}}{C_{ox}}$$

$$V_T = V_{FB} + 2\phi_F + \frac{\sqrt{2qN_A\varepsilon_{Si}(2\phi_F)}}{C_{ox}}$$

- **For n-type Si:**

$$V_T = V_{FB} + 2\phi_F - \frac{\sqrt{2qN_D\varepsilon_{Si}|2\phi_F|}}{C_{ox}}$$

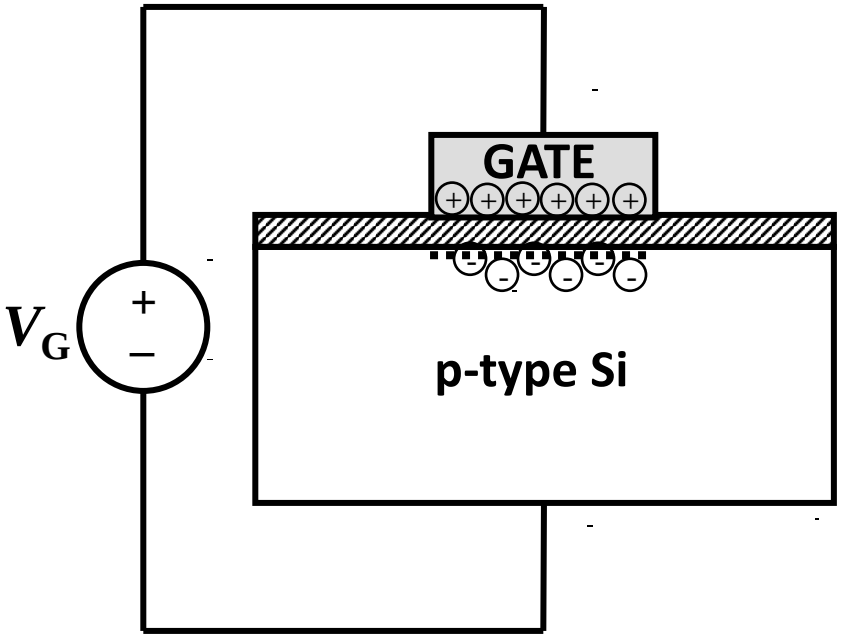


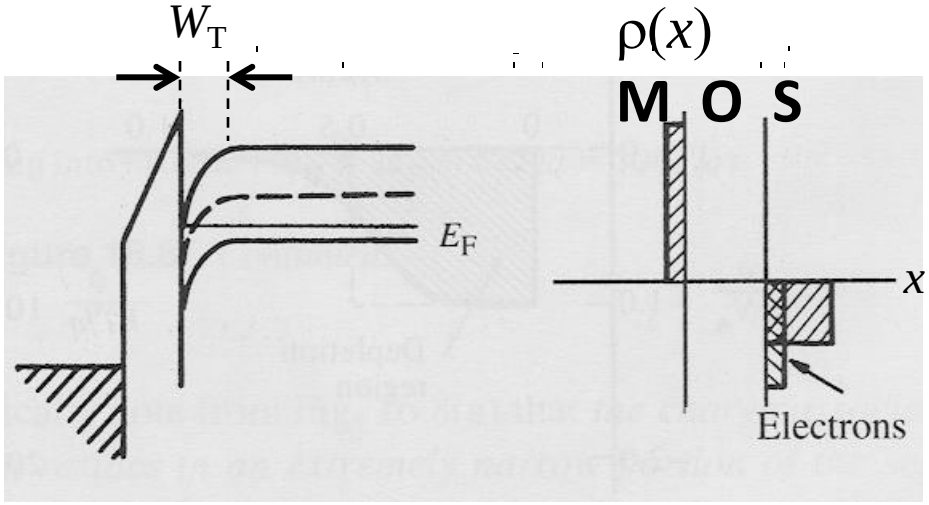
- $|V_T|$ increases with Si doping
- $|V_T|$ increases with increasing x_{ox}

C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 5-8

Strong Inversion (p-type Si)

As V_G is increased above V_T, the negative charge in the Si is increased by adding mobile electrons (rather than by depleting the Si more deeply), so **the depletion width remains ~constant at $W = W_T$**





Significant density of mobile electrons at surface (surface is n-type)

$$\phi_S \cong 2\phi_F$$

$$W \cong W_T = \sqrt{\frac{2\varepsilon_{si}(2\phi_F)}{qN_A}}$$

R. F. Pierret, *Semiconductor Device Fundamentals*, p. 575

Inversion Layer Charge Density (p-type Si)

$$V_G = V_{FB} + \phi_S + V_{ox}$$

$$= V_{FB} + 2\phi_F - \frac{(Q_{dep} + Q_{inv})}{C_{ox}}$$

Charge density within the Si = $Q_{\text{dep}} + Q_{\text{inv}}$

$$= V_{FB} + 2\phi_F + \frac{\sqrt{2qN_A\varepsilon_s(2\phi_F)}}{C_{ox}} - \frac{Q_{inv}}{C_{ox}}$$

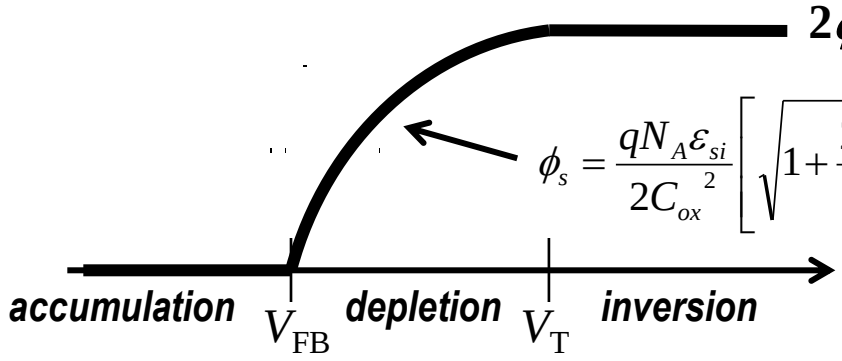
$$= V_T - \frac{Q_{inv}}{C_{ox}}$$

$$\therefore \quad \boxed{Q_{inv} = -C_{ox}(V_G - V_T)}$$

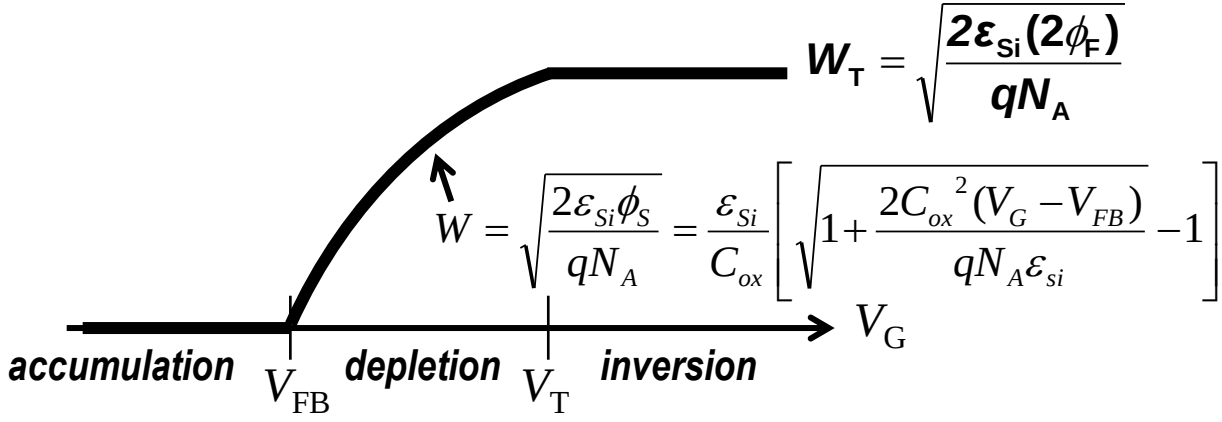
Units: C/cm^2

ϕ_S and W *vs.* V_G (p-type Si)

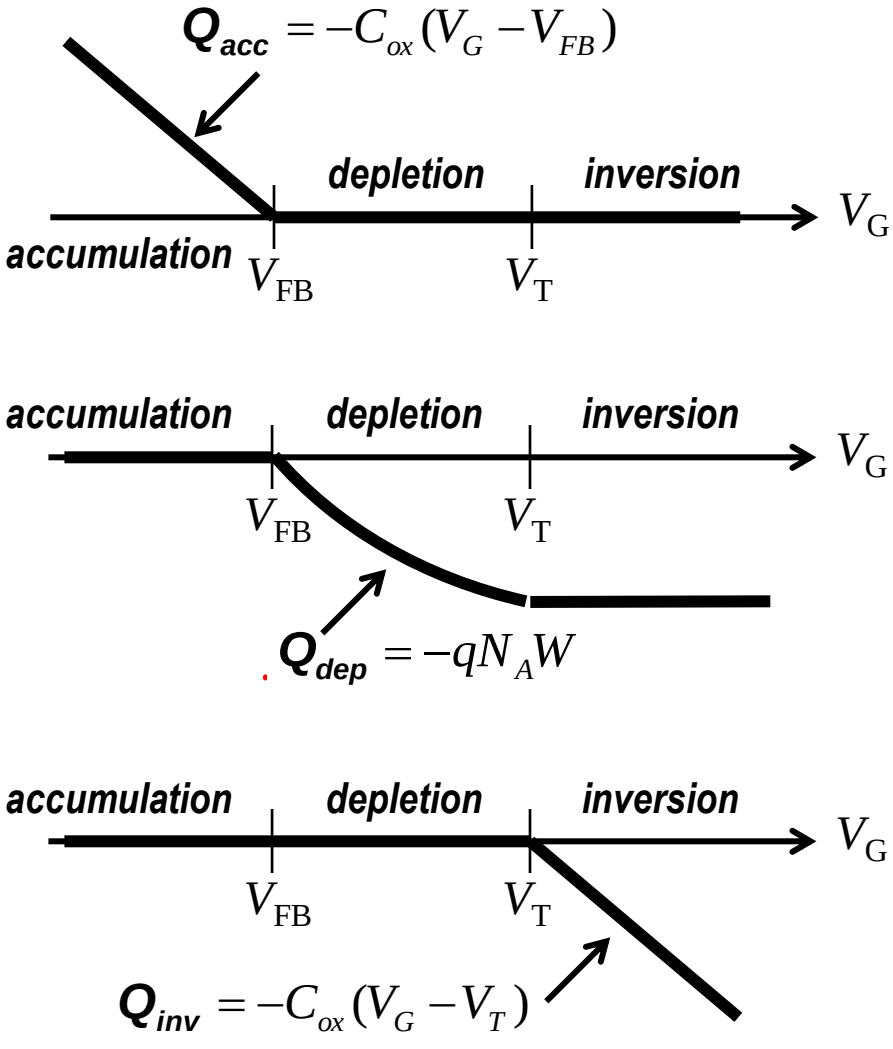
ϕ_S:



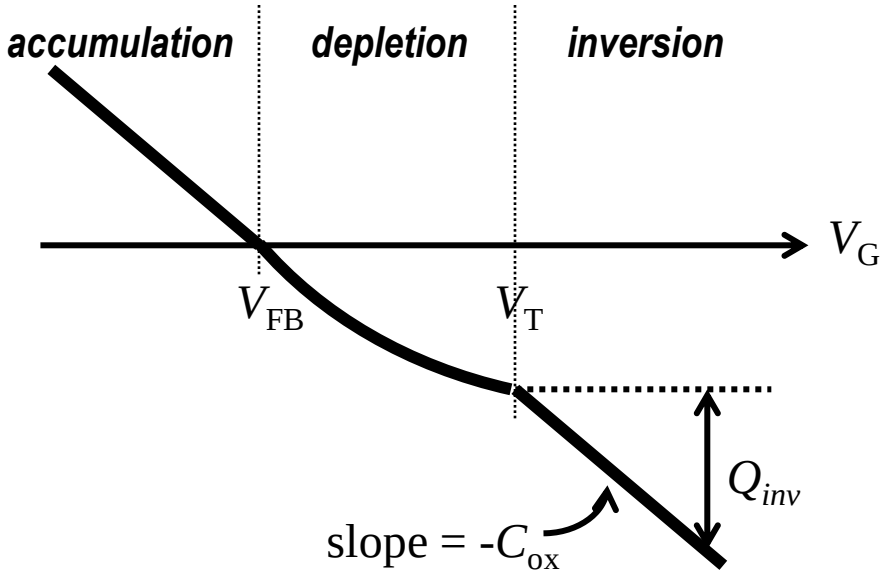
W:



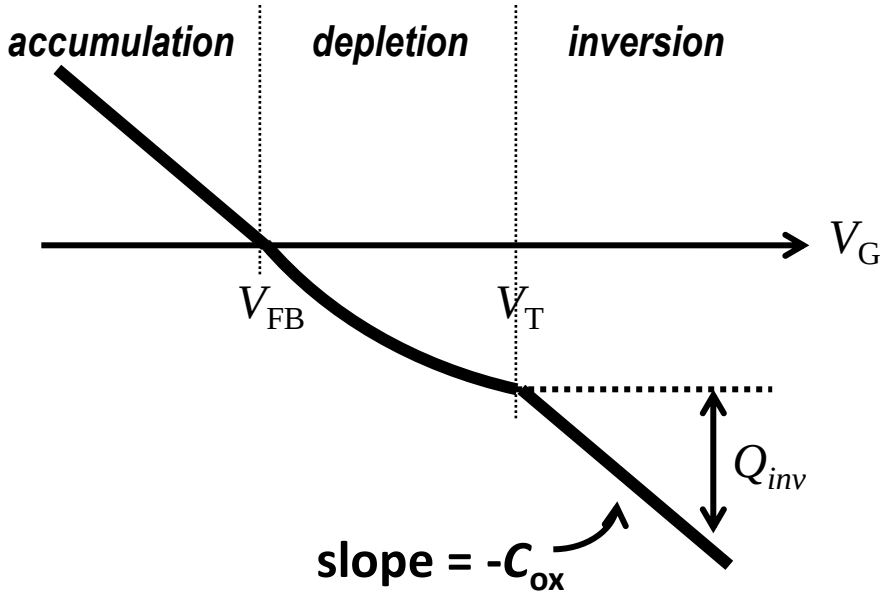
Total Charge Density in Si, Q_s (p-type Si)



$$Q_s = Q_{acc} + Q_{dep} + Q_{inv}$$

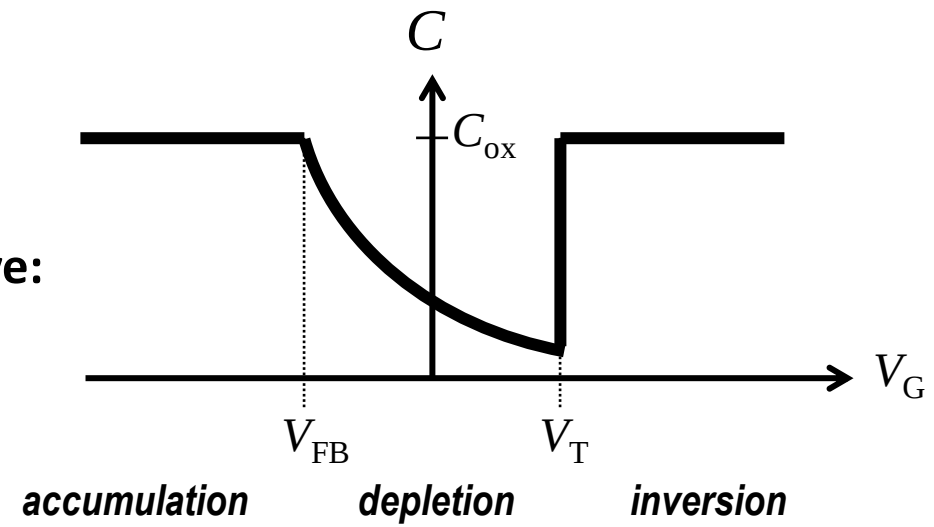


MOS *C-V* Characteristics (p-type Si)

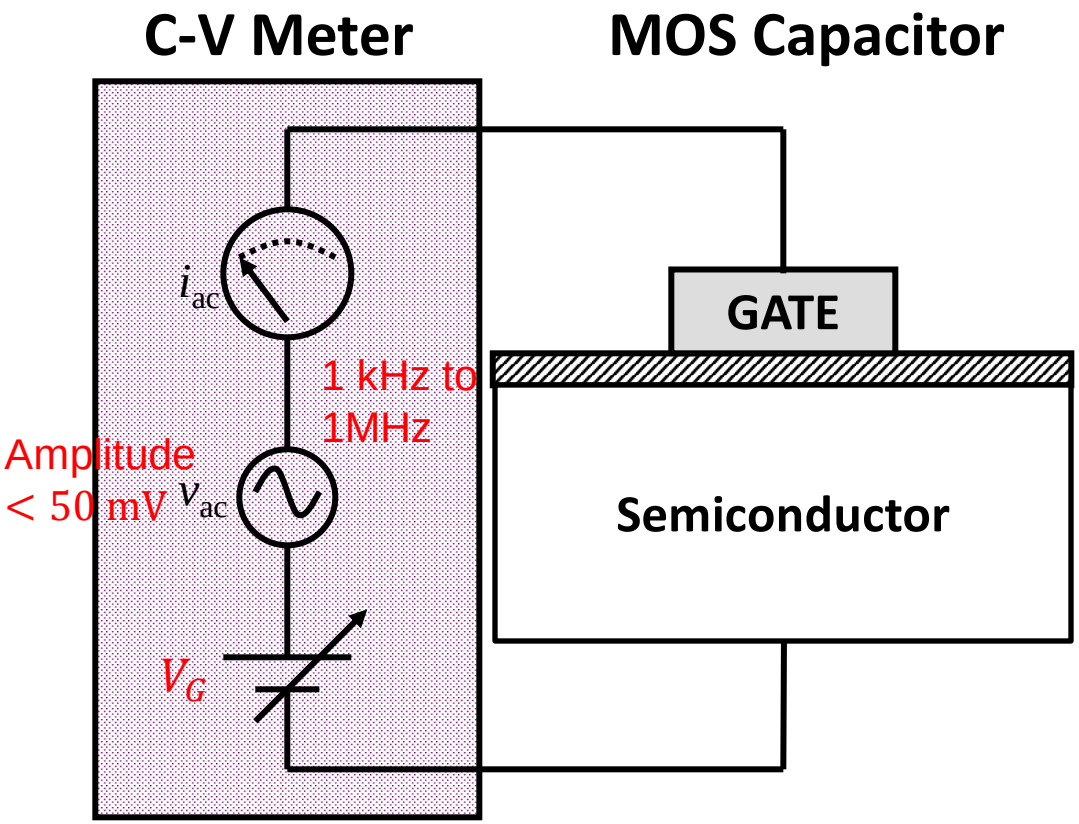


$$C = \left| \frac{dQ_s}{dV_G} \right|$$

Ideal *C-V* curve:



MOS Capacitance Measurement

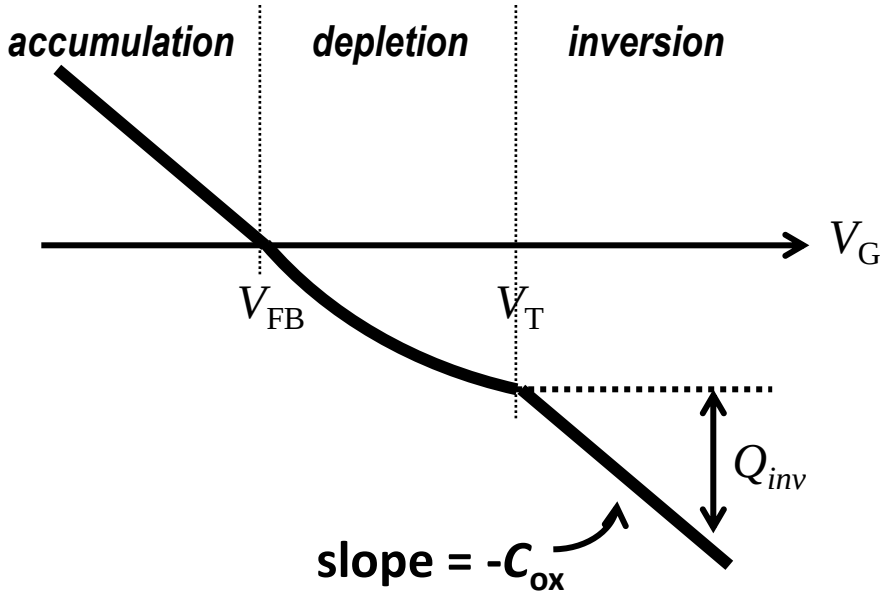


- V_G is scanned slowly (0.1V/s)
- Capacitive current due to v_{ac} is measured

$$i_{ac} = C\frac{dv_{ac}}{dt}$$

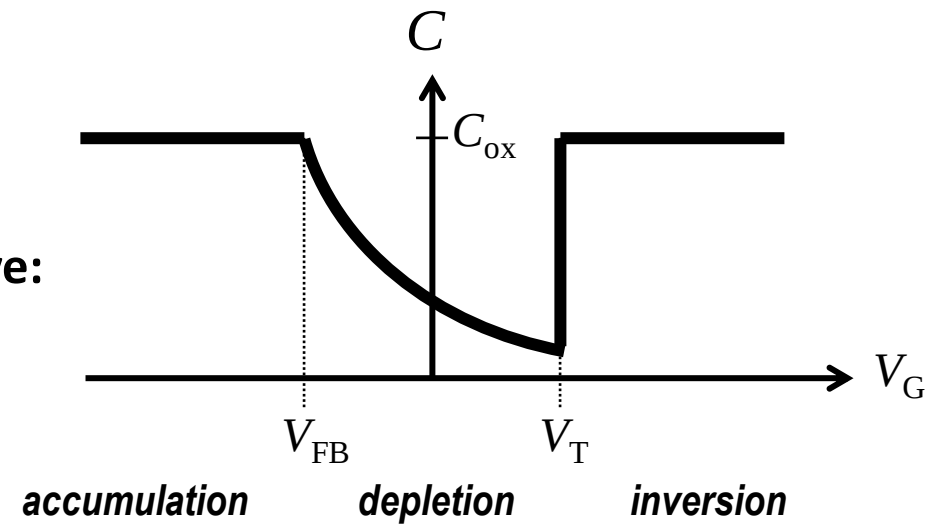
$$C = \left|\frac{dQ_{GATE}}{dV_G}\right| = \left|\frac{dQ_s}{dV_G}\right|$$

MOS *C-V* Characteristics (p-type Si)



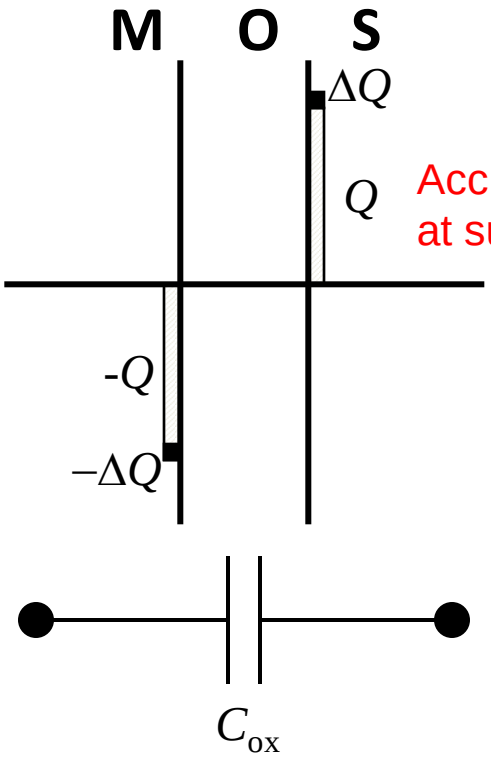
$$C = \left| \frac{dQ_s}{dV_G} \right|$$

Ideal *C-V* curve:



Capacitance in Accumulation (p-type Si)

- **As the gate voltage is varied, incremental charge is added** (or subtracted) **to** (or from) **the gate and substrate.**
  - The incremental charges are separated by the gate oxide.



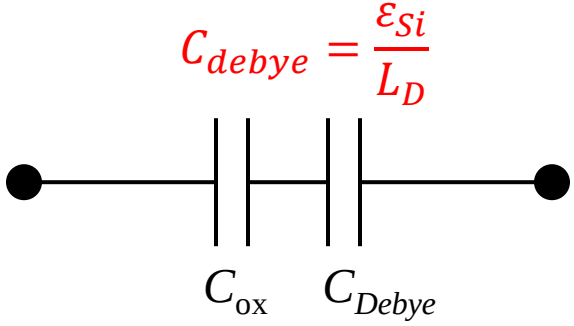
Accumulation layer of holes at surface of p-type Si

$$C = \left| \frac{dQ_{acc}}{dV_G} \right| = C_{ox}$$

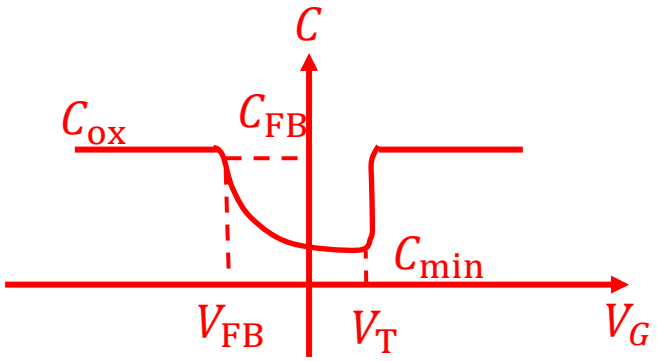
Flat-Band Capacitance (p-type Si)

- **At the flat-band condition, variations in V_G give rise to the addition/subtraction of incremental charge in the substrate, at a depth L_D**
  - L_D is the "extrinsic Debye Length," a characteristic screening distance, or the distance where the electric field emanating from a perturbing charge falls off by a factor of $1/e$

$$L_D \equiv \sqrt{\frac{\varepsilon_{Si} kT}{q^2 N_A}}$$



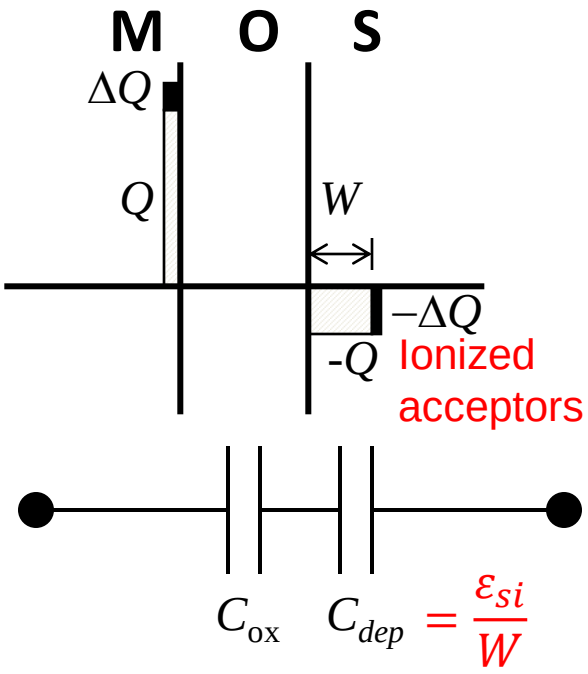
$$\frac{1}{C_{FB}} = \frac{1}{C_{ox}} + \frac{L_D}{\varepsilon_{Si}}$$



Capacitance in Depletion (p-type Si)

- **As the gate voltage is varied, the depletion width varies.**

  → Incremental charge is effectively added/subtracted at a depth *W* in the substrate.



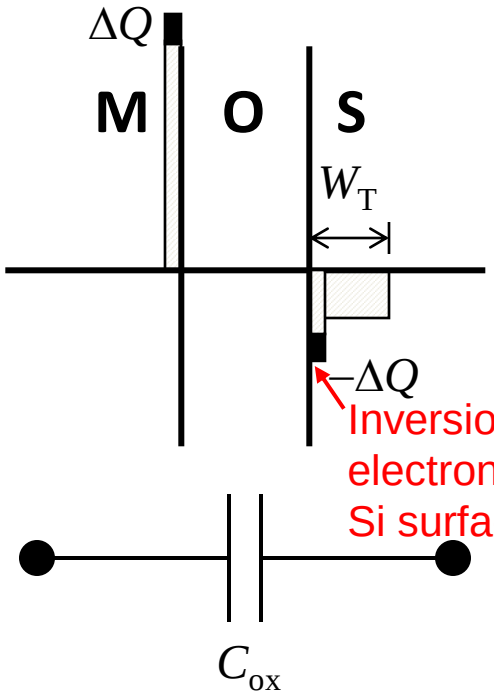
$$C = \left|\frac{dQ_{dep}}{dV_G}\right| = \sqrt{\frac{1}{C_{ox}^{\ 2}} + \frac{2(V_G - V_{FB})}{qN_A\varepsilon_{Si}}}$$

$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_{dep}} = \frac{1}{C_{ox}} + \frac{W}{\varepsilon_{Si}}$$

Capacitance in Inversion (p-type Si)

<u>CASE 1</u>: Inversion-layer charge *can* be supplied/removed quickly enough to respond to changes in gate voltage.

→ Incremental charge is effectively added/subtracted at the surface of the substrate.



By thermal generation within the depletion region

Time required to build inversion-layer charge = $2N_A\tau_o/n_i$, where

τ_o = minority-carrier lifetime at surface

τ_o typically 1ns to 1μs so time required to build inversion layer is ~0.1s

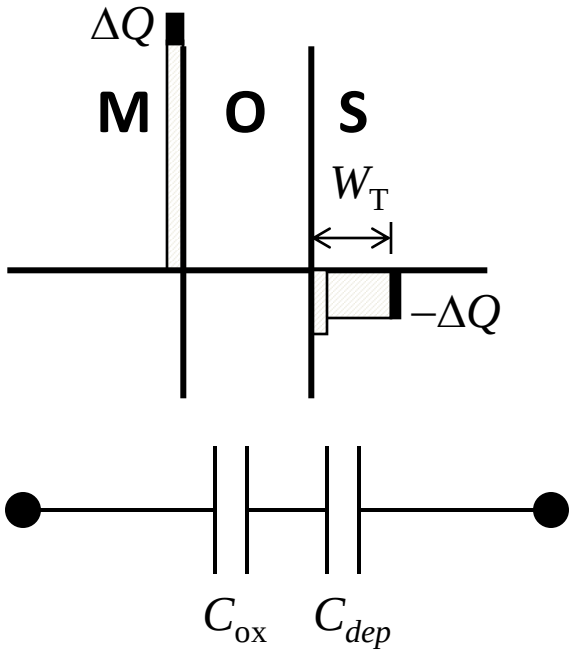
$$C = \left|\frac{dQ_{inv}}{dV_G}\right| = C_{ox}$$

Thermal generation is not fast enough to respond to high frequency AC voltage signal

Capacitance in Inversion (p-type Si)

<u>CASE 2</u>: Inversion-layer charge *cannot* be supplied/removed quickly enough to respond to changes in gate voltage.

→ Incremental charge is effectively added/subtracted at a depth W_T in the substrate.



$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_{dep}}$$

$$= \frac{1}{C_{ox}} + \frac{W_T}{\varepsilon_{Si}}$$

$$= \frac{1}{C_{ox}} + \sqrt{\frac{2(2\phi_F)}{qN_A\varepsilon_{Si}}} \equiv \frac{1}{C_{\min}}$$

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- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
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